OTAZ III: The Gibbs Ledger — Exact Aggregation, a Sharp Impossibility Theorem, and Certified Error Budgets for Scored-Record Fusion

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Status. This document is a preprint. It has not been peer reviewed, it has not been submitted to a journal, and no claim of acceptance or review is made anywhere in it. A one-page novelty assessment accompanies the deposit and states, item by item, what is classical, what is a repackaging, and what is offered as new and checkable. All code, data and figures needed to re-derive every number in the text are included in the accompanying repository under the PolyForm Noncommercial 1.0.0 licence; the text is under CC BY-NC 4.0.

Abstract. A scored record is a pair: a value and a score. Fusing a collection of such records into one pair — a Gibbs-weighted mean of the values together with a scaled log-sum-exp of the scores — is one of the oldest operations in statistics and one of the most frequently re-derived in engineering. This paper treats the pair as an accounting object and asks three questions that the usual treatments leave implicit: what algebra the operation has, what a distributed implementation must carry between shards, and what guarantees survive when the inputs are estimated rather than known.

The algebra is settled first. The records form a commutative monoid whose operation is associative but not idempotent — duplicating a record adds exactly $\beta^{-1}\ln 2$ to the score — and which is not homogeneous in the values, so the operation is not the first slot of a bilinear form. The binary rule generates the n-ary one, so aggregation is order-independent, and the pair (mass, mass-weighted sum) is a sufficient statistic, which makes tree-structured merging exact.

The distributed question is then answered negatively, and this is the central contribution. If each shard reports only its value — however many records it contains — the set of global values compatible with the reports is the whole convex hull of the shard values, and no merge rule can do better, in the worst case, than half the range of those values. The bound is sharp, the minimax rule is the midpoint, and equal weighting is never better. Two datasets with identical per-shard values and identical record counts can differ in their global value by an amount proportional to the mass ratio the summary discarded; a worked example gives a gap of 1.545 on a range of 10. If the masses are carried but rounded to relative precision δ, the merged value is accurate to $D\delta/(1-\delta)$ whatever the number of shards; both bounds are proved and measured.

On the positive side I prove that block-wise evaluation of a pipeline loses exactly β^{-1} times the entropy of the block-mass distribution, bounded by $\beta^{-1}\ln k$ and attained by equal masses; that the loss obeys a Shannon chain rule across levels; that the logit parametrisation cannot amplify an L_1 error of the weights; and that sampling and truncation budgets compose into one certified interval which held in all 300 random pipelines tried. A closing technical section fixes the boundary of the quantum reading: the weight vector is the diagonal of a density matrix, dephasing contracts the trace distance so the classical computation is a lower bound, a Helstrom floor follows, and two orthogonal states with identical diagonals show that no converse exists.

Sixteen seeded experiments, 52 unit tests and a 56-check independent verifier support the text; the negative results, including a bootstrap that under-covers and a certificate that saturates at small sample sizes, are reported in the main text rather than in a footnote. The novelty assessment is deliberately conservative: the ingredients are classical and their sources are named, and the one result offered as new is the sharp impossibility statement for value-only summaries together with its quantisation refinement.

Index Terms. Gibbs distribution, log-sum-exp, softmax, fusion, mergeable summaries, distributed aggregation, minimax lower bound, quantisation, entropy, certified error bounds, trace distance, dephasing, reproducible research.

I. Introduction

A. The object

Most systems that produce a quantity also produce a claim about how much that quantity should be trusted. A weather model emits a temperature and a verification score; a learned ranker emits a probability and a calibration measure; a sensor emits a reading and a signal-to-noise ratio; a Monte-Carlo estimator emits a sample and a log-weight. When such records must be combined, the combination has to respect both halves of the pair, and the classical answer is a Gibbs average: exponentiate the scores, normalise, take the weighted mean of the values, and report the scaled logarithm of the partition sum as the aggregate's own score.

$$m_i = e^{\beta w_i}, \quad p_i = m_i/Z, \quad Z = \sum_i m_i, \quad V = \sum_i p_i v_i, \quad W = \beta^{-1}\ln Z. \tag{1}$$

I call m_i the mass of record i, Z the partition sum, V the fused value and W the fused score, and the triple (Z, V, W) the ledger of an aggregate. The word is deliberate. A ledger is not a summary of what happened; it is the set of numbers one must keep in order to answer later questions exactly, and the difference between those two things is the subject of Section IV.

The operation appears under many names: exponential weighting, logarithmic pooling, log-mean-exp pooling, soft gating in a mixture of experts, Bayesian model averaging over a finite model set, likelihood-ratio combination, the recursive update of a particle filter, and the partition function of a Gibbs distribution with a fixed

sufficient statistic. Each community has re-derived the two-line algebra above and each has, reasonably, stopped there, because the two lines are easy and the applications are pressing. The cost of stopping there is that the interesting questions are never asked. Three of them organise this paper.

B. Three questions

What algebra does the operation have? Texts state that the fused score is a log-sum-exp and the fused value a weighted mean, and then move on. But the properties engineers rely on — may I merge in any order? may I merge two partial results without revisiting the inputs? what happens if a record arrives twice? — are algebraic properties, and they are not all visible in the formula. Duplication is not neutral: a record that arrives twice is not the same as a record that arrives once, because the partition sum counts it twice while the value does not move, and the size of the effect is exactly $\beta^{-1}\ln 2$. Section III proves the algebra in full, including the parts that fail.

What must a distributed implementation carry between shards? This is the question with the sharpest answer, and I regard it as the contribution of the paper. The tempting interface is to send the value: each shard computes its own fused value and reports it. The correct interface is to send the mass as well. Section IV shows that the difference is not a matter of taste. With values alone, the set of global values still compatible with the reports is the entire convex hull of the reported values, so a merge rule can be wrong by half the range of those values no matter how clever it is; the midpoint is minimax and attains that bound exactly. Two datasets whose shards agree in value and in record count but differ in mass can produce global values differing by 1.5452 on a range of 10 — about a sixth of the range, from two datasets no value-only rule can tell apart. If the mass is carried but quantised, the loss is $D\delta/(1-\delta)$, and that is sharp too.

What survives when the inputs are estimated? Production inputs are never known: weights come from finitely many samples, some records are dropped for cost, and some aggregates are computed block by block. Each introduces an error, and a system that reports a number without an interval around it reports an illusion. Sections VI and VII supply the three statements that make an interval possible: the exact cost of coarse evaluation, an error non-amplification result for the logit parametrisation, and a composition theorem that turns sampling and truncation into one certified bound. Section VIII takes the last step and asks what the same construction means when the weights come from a quantum device; the answer is precise and one-directional.

C. What this paper does not claim

The mathematics is built from classical parts: the Gibbs variational principle, the entropy bound, the pinching inequality, the Helstrom formula, elementary minimax reasoning. Section IX lists the components against their sources in one table and states, for each, whether it is classical, repackaged or offered as new. Exactly one item is offered as new: the impossibility statement of Section IV with its sharp constant and its quantisation refinement, in the setting of a fusion operator that carries a score alongside a value. Even there, the style of argument is standard minimax theory and the framing owes an obvious debt to the mergeable-summaries literature; what I could not find is the statement itself for this operator. The one-page novelty assessment that accompanies the deposit repeats this in a form a reader can attack item by item.

Two conventions. A result is not called new unless a proof is given and the nearest prior art is named. And no number appears in the text that is not produced by the accompanying code from a stated seed and stored in a named file; when the code and the prose disagree, the code wins and the prose is a bug.

D. Contributions

1. **Algebra.** The records form a commutative monoid, associative and exactly mergeable, with an explicit two-number sufficient statistic; the operation is not idempotent (duplication adds $\beta^{-1}\ln 2$ exactly) and not homogeneous, so it is not the first slot of a bilinear form (Theorems 1–2).
2. **Impossibility.** Value-only summaries leave the merged value undetermined on the full convex hull of the shard values (Theorem 3); every merge rule has worst-case error at least half the range, and the midpoint attains it (Theorem 4); a ledger rounded to relative precision δ merges to within $D\delta/(1-\delta)$, independently of the number of shards, with the constant not claimed sharp (Theorem 5).
3. **Accounting.** Block-wise evaluation loses exactly $\beta^{-1}\times$the entropy of the block masses, bounded by $\beta^{-1}\ln k$, attained by equal masses, and obeying a Shannon chain rule across levels (Theorems 7–8).
4. **Certificates.** The logit parametrisation cannot amplify an L_1 error (Theorem 9), and sampling plus truncation compose into a single certified interval (Theorem 10).
5. **Quantum boundary.** The theory is the commuting special case; dephasing contracts the trace distance so the classical computation is a lower bound (Theorem 11), a Helstrom floor follows (Corollary 1), and two orthogonal states with identical diagonals rule out a converse (Theorem 12).
6. **Reproducibility.** Sixteen seeded experiments, 52 unit tests, a 56-check verifier that does not import the library it checks, a checksum manifest and stored results (Section XIII).

II. Setting, notation and related work

A. Notation

A record is a pair (v, w) with value $v \in \mathbb{R}^d$ and score $w \in \mathbb{R}$. The parameter β is a finite non-zero real; by convention $\beta > 0$ sharpens the weights toward high scores and $\beta < 0$ toward low ones, and several statements below fix a sign because an inequality reverses when β^{-1} changes sign. D is the diameter of the values under the norm in use. Total variation is $\mathrm{TV}(p, q) = \tfrac{1}{2}\|p-q\|_1$. Trace distance is $\tfrac{1}{2}\|\rho-\sigma\|_1$; the two agree for diagonal states.

B. Numerical realisation

Equation (1) must not be evaluated as printed. The exponentials overflow in double precision for βw beyond roughly 709 and underflow below roughly -745, and both failures are silent: the sum becomes zero or infinite and the subsequent division produces a not-a-number that propagates without a warning in most array libraries. The remedy is the max-shifted form $\beta^{-1}(c + \ln\sum_i e^{\beta w_i - c})$ with $c = \max_i \beta w_i$, which leaves the weights

unchanged and shifts the score by the known amount $\beta^{-1}c$. The shifted form is what the accompanying library implements, and the test suite includes scores of order 800 so that a regression to the naive form fails loudly. The error analysis of this primitive, including bounds on the relative error of the result, is due to Blanchard, Higham and Higham [8]; the one-pass streaming version to Milakov and Gimelshein [9].

The natural object for a distributed implementation is not the pair but the triple (Z, S, n) with $S = \sum_i m_i v_i$, because Z and S are additive and the pair is recovered by division and logarithm. Section III shows that this is forced rather than convenient; Section IV shows what happens when an implementation drops Z.

C. Where the object appears

Forecast combination. Combining predictive distributions under a logarithmic score reduces to a score-weighted average whose weights are softmax in the accumulated scores; the fifty-year review of Wang, Hyndman, Li and Kang [12] documents both the practice and the strength of the equal-weight baseline, which is why Section IV states its comparison against equal weighting specifically.

Mixtures and products of experts. Soft gating computes $g_i = \mathrm{softmax}(s_i)$ and returns $\sum_i g_i e_i(x)$ [20], which is (1) with the gate logit in the role of the score. Product-of-experts models use the same log-sum-exp arithmetic in the log domain [21], which is the same function read as a smooth maximum [3].

Sequential Monte Carlo. Particle filters carry log-weights and combine them by exactly this recursion; the fused score is the estimated log-evidence [22].

Distributed estimation. When devices cannot share raw data they share summaries, and which summaries compose is the subject of the mergeable-summaries literature [16, 17]; linear sketches are trivially mergeable. Section III shows that the ledger is a linear sketch of the required kind, and Section IV shows that anything smaller is not.

Quantum state estimation. A weight vector is a diagonal density matrix, and what a classical summary of a quantum state can certify is the classical-versus-quantum question of Section VIII [4, 5, 6].

D. Related theory

Three bodies of work bound what this paper can claim. The exponential-family literature establishes that the natural sufficient statistic of a Gibbs family is additive and of fixed dimension [18], which is the content of Theorem 1(v) and of the merge procedure. The mergeability framework of Agarwal et al. [16], with the sketching literature [17] alongside, formalises when a summary can be merged while preserving an error and size guarantee, and proves both positive and negative results for concrete summaries. Communication-constrained minimax theory [19] gives lower bounds on estimation error as a function of a bit budget, via metric entropy. The result of Section IV sits between the last two: the constraint is not the number of bits but the shape of the summary, and because that shape is so restrictive the bound is exact rather than order-optimal, with an explicit attaining construction.

III. The algebra of scored-record fusion

A. The two slots and what they are called

Write a record as (v, w) and define the fusion of two records by

$$(v_A, w_A) \bowtie_\beta (v_B, w_B) = \big((m_A v_A + m_B v_B)/(m_A + m_B) \;\big|\; \beta^{-1}\ln(m_A + m_B) \big), \tag{2}$$

with $m_A = e^{\beta w_A}$ and similarly for B. The left slot is a mass-weighted mean of values; the right slot is a scaled log-sum-exp of masses. Everything in this paper follows from two facts about these slots: the left slot is unchanged when the same constant is added to every score, and the right slot is additive in masses. It is worth naming the notation that (2) inherits. The vertical bar and the angle brackets come from the Dirac bracket, where $\langle\varphi|\psi\rangle$ is a sesquilinear form and, when the arguments are drawn from the same space, an inner product. That reading is tempting and wrong, and the failure is elementary enough to state as a lemma rather than a remark.

B. The bracket is not an inner product

Lemma 1 (no bilinearity). Let $F(v_A, w_A; v_B, w_B)$ be the left slot of (2). For $\lambda \neq 0$ and $w_A \neq w_B$, $F(\lambda v_A, w_A; v_B, w_B) = \lambda F(v_A, w_A; v_B, w_B)$ fails unless $\lambda = 1$ or $v_B = 0$. With $(v_A, w_A) = (1, 0.5)$, $(v_B, w_B) = (3, 1.5)$, $\beta = 1$ and $\lambda = 2$, the two sides differ by 2.1932.

Proof. Put $\alpha = m_A/(m_A + m_B)$; then $F = \alpha v_A + (1-\alpha)v_B$. Scaling v_A by λ gives $\lambda\alpha v_A + (1-\alpha)v_B$, while λF gives $\lambda\alpha v_A + \lambda(1-\alpha)v_B$. These agree for every λ only if $(1-\alpha)v_B = 0$. The numerical discrepancy is stored in results/e13_binary_algebra.json. □

The consequence is not merely notational. A reader who treats the bracket as an inner product will expect to be able to superpose, to normalise, and to project, and none of those operations is available. In the remainder of the paper I therefore write the two slots explicitly as (V, W) and never use the bracket again. What the operation does have is a clean algebraic description, and it is worth more than the geometric one it lacks.

C. The ledger monoid

> **Theorem 1 (the ledger).**
> Fix β and consider records (v, w) with $v \in \mathbb{R}^d$. Then:
>
> (i) $\bowtie_\beta$ is commutative;
>
> (ii) $\bowtie_\beta$ is associative, and the aggregate of a family depends only on the multiset of records, not on the order or the bracketing of the merges;
>
> (iii) $\bowtie_\beta$ is not idempotent: $(v, w) \bowtie_\beta (v, w) = (v, w + \beta^{-1}\ln 2)$, so a k-fold duplicate shifts the score by $\beta^{-1}\ln k$ and never changes the value;
>
> (iv) there is no identity element among finite records and the operation is not invertible;
>
> (v) the map $\sigma(v, w) = (m, mv)$ is a sufficient statistic: Z and $S = \sum_i m_i v_i$ add over merges, and (V, W) is recovered from (Z, S) by division and logarithm;
>
> (vi) the pair is invariant to a common score shift, $w_i \to w_i + c$, which leaves V and every weight unchanged and shifts W by c.

Proof. Commutativity is self-evident in (2). For associativity, observe that the right slot of $A \bowtie B$ is $\beta^{-1}\ln(m_A+m_B)$, so the mass of an aggregate is the sum

of the masses, and the left slot is the mass-weighted mean; merging the aggregate with C therefore yields mass $m_A+m_B+m_C$ and the three-record weighted mean, which is symmetric. Formally, in the coordinates of (v) the fusion is vector addition and the pair is a function of the sum, so associativity and commutativity of the operation follow from those of addition. For (iii), set $v_A = v_B = v$: the left slot is v and the right slot is $\beta^{-1}\ln(2e^{\beta w}) = w + \beta^{-1}\ln 2$; induction gives the k-fold claim. For (iv), an identity would need mass zero, which no finite record has, and σ is many-to-one, so no inverse exists. Part (vi) is immediate from $m_i \to e^{\beta c}m_i$. The full argument for (ii) and (v) is in Appendix A.1. □

Four consequences are worth spelling out, because each corresponds to a failure mode seen in practice.

Duplication. Part (iii) says that two identical records are not one record. If a pipeline de-duplicates before fusing, it changes the score by $\beta^{-1}\ln 2$ per doubling, and through the score it changes every downstream weight and every decision threshold that consumes the score. The error is large when β is small: 0.69 nats at $\beta = 1$, 6.93 at $\beta = 0.1$. Because the size is known, the defect is fixable; the experiment confirms the prediction to 2.8×10^{-17} across three temperatures.

Order. Part (ii) is why a distributed merge can be arranged in any tree. A reviewer might ask why associativity needs proof for an operation that is a weighted average. The answer is that the two slots interact: the right slot of one fusion is the score of the next, so associativity of the pair is not the associativity of a mean, and it fails for the natural alternative in which the score is carried along unchanged. Experiment E16 merges up to 10,000 records in blocks of 1, 10 and 100 with random tree structures and finds the tree result equal to the n-ary result to a relative 3.7×10^{-16}.

Distribution. Part (v) says an aggregate is summarised by two numbers per coordinate, total mass and mass-weighted sum, both additive. A pipeline therefore merges partial aggregates exactly, in any tree, without revisiting the leaves. This is the property the mergeable-summaries literature calls mergeability [16, 17], and the ledger is a linear sketch in exactly their sense. Nothing here is new; what Section IV adds is the converse direction, namely the price of a summary that is not of this shape.

Gauge. Part (vi) says the score has a free additive constant. That is why scores from two sources cannot be compared until a common offset is fixed, and why a system that merges scores produced under different calibrations is silently wrong: the weights are invariant to the offset but the merged score is not, and the merged score is what the next merge consumes.

D. The n-ary form and order independence

Theorem 2 (the binary law generates the n-ary form).

For records $(v_1, w_1), \ldots, (v_n, w_n)$ with $n \geq 2$, every binary bracketing of $\bowtie_\beta$ over the n records yields the pair

$$V = \sum_i p_i v_i, \quad W = \beta^{-1} \ln \sum_i e^{\beta w_i}, \quad p_i = e^{\beta w_i}/\sum_j e^{\beta w_j}. \tag{3}$$

Consequently a binary implementation, an n-ary implementation and a tree-structured merge produce the same pair up to floating-point rounding.

Proof. Induction on n; the case $n = 2$ is (2). Suppose the claim holds for a subset S of size $n-1$, with summary mass $\sum_{i\in S} m_i$ and numerator $\sum_{i\in S} m_i v_i$. Fusing with the remaining record multiplies the mass by $e^{\beta w_n}$ and adds the corresponding numerator, which is the n-ary expression. Associativity (Theorem 1(ii)) removes the dependence on bracketing. □

Numerically, (3) is not evaluated as printed, for the reason given in Section II.B; the library shifts by the maximum score. The identity also settles a design question that is otherwise arbitrary: an aggregate that carries Z can be merged with another aggregate without knowing how many records either contains, because Z is the sufficient statistic and $e^{\beta W}$ recovers it. An aggregate that carries only V cannot be merged at all without an assumption. That asymmetry is Theorem 1(v) read as an interface, and Section IV quantifies the assumption.

Table 1. Algebraic checklist for the operation of (2), with the measurement that produced each verdict. Source: results/e13_binary_algebra.json, results/e16_tree_merge.json.

Property	Holds	Evidence
Commutative	yes	max defect 5.6×10^{-16}, 200 random pairs
Associative, both slots	yes	max gap to the n-ary form 8.9×10^{-16}, 300 triples
Tree merging equals n-ary merging	yes	relative gap 3.7×10^{-16} up to 10^4 records
Idempotent	**no**	offset $\beta^{-1}\ln 2$, residual ≤ 2.8×10^{-17}
Homogeneous in the value	**no**	discrepancy 2.19 at $\lambda = 2$
Has an identity element	**no**	would require zero mass
Invertible	**no**	the sufficient statistic is many-to-one
Invariant to a common score shift	yes (value), shift (score)	Theorem 1(vi)

IV. The ledger is necessary: an impossibility theorem and its sharp constant

A. The interface question

Suppose a pipeline is partitioned across k shards. Each shard holds a collection of scored records whose union is the full collection. Two designs are possible for the interface between a shard and the merge service. In the first, the shard reports the pair (V_B, W_B) computed from its own records, and the merge service combines the reported pairs using (2). In the second, the shard reports only its value V_B — possibly with the number of records n_B — and the merge service does what it can.

The first design is exact by Theorem 1: the pair determines the mass $Z_B = e^{\beta W_B}$ and the numerator $S_B = Z_B V_B$, both additive, so the merge is a two-line computation with no error at all. The second design is what a cost-conscious implementation drifts toward, because a value is one number and a pair is two, and because a downstream consumer often wants only the value. It is also what happens by accident whenever a shard computes its own average, reports it, and the merge service averages the reports.

This section settles what the second design costs. The answer has three parts: the set of global values that remain possible is exactly the convex hull of the reported

values; every merge rule has worst-case error at least the radius of that hull, which in one dimension is half the range, and the midpoint attains it; and if the mass is carried but rounded, the error is bounded by Dδ/(1−δ). None of these statements is deep. Together they convert a stylistic choice into a quantitative one, which is what an architect needs in order to decide how much of the ledger to keep.

B. What a value-only summary can and cannot determine

Theorem 3 (the reachable set is the hull).
Fix β and let a partition into k blocks have reported values $V_1, \ldots, V_k \in \mathbb{R}^d$. Consider any merge rule Γ whose input is the collection of pairs $(V_B, w_B^{(\text{inside})})$ in the sense that it may use anything computable from within block B except the block's normalising mass. Then the set of global values compatible with what Γ sees is exactly

$$S = \operatorname{conv}\{ V_1, \ldots, V_k \}, \tag{4}$$

the convex hull of the reported values. In particular the number of records per block, and any function of the within-block weights, adds nothing.

Proof. That $S \subseteq \operatorname{conv}\{V_B\}$ is immediate: the global value is a convex combination of the block values with weights $\pi_B = Z_B/Z \geq 0$ summing to one.

For the reverse inclusion, take any π in the simplex. It is enough to exhibit, for each block B, a block of records whose reported value is V_B and whose mass is any prescribed positive number M_B. Take a single record (V_B, w_B) with $w_B = \beta^{-1} \ln M_B$: its value is V_B and its mass is M_B. Then the global value is $\sum_B \pi_B V_B$ for $\pi_B = M_B/\sum_C M_C$, and since the M_B are free the weights range over the whole simplex. The same construction works when blocks must contain at least n_B records, because adding records with the block value V_B changes neither the value nor the sensitivity to the free mass. Appendix A.2 gives the construction in the multi-record case. □

The mechanism deserves a name, because it recurs. Shifting every score inside a block by the same constant c multiplies that block's mass by $e^{\beta c}$ and leaves every within-block weight, hence the block's value, unchanged (Theorem 1(vi)). A value-only summary is therefore blind to a quantity that the merge needs, and blind in a way that cannot be repaired by any local computation: the information was never collected.

A concrete pair. Let block A contain two records with values 0 and 10 and scores 0 and 0, and let block A′ contain two records with the same two values and scores 3 and 3. Both blocks report $V_A = 5.0$ and $n_A = 2$; their masses are 2.0 and 40.17. Merge each with a third block holding a single record of value 0 and score 0. The two global values are **3.3333** and **4.8786** — a difference of **1.5452** on a value range of 10, produced by two datasets that are indistinguishable to any merge rule that looks at values and counts. The numbers are regenerated by experiment E12 and stored in results/e12_impossibility.json.

The pair also shows why a pipeline cannot repair the omission downstream. Once the mass is gone, two different datasets are consistent with the same report, so no post-hoc adjustment, calibration or reweighting can separate them: a rule that handles one of the two correctly must mishandle the other. That is the sense in which the ledger is necessary rather than merely advisable.

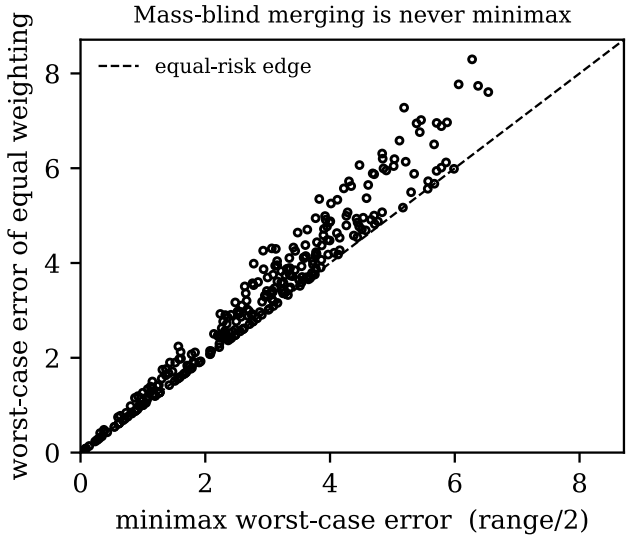


Figure 1. Worst-case error of equal weighting against the minimax worst-case error, over 300 instances with two to six blocks and values drawn from a normal distribution. Every point lies on or above the diagonal: the equal-weight merge is never better than the minimax rule, and it coincides with it only when the mean happens to be the midpoint of the extremes. The horizontal axis is the sharp constant of Theorem 4, namely half the range. Source: results/e12_impossibility.json.

C. The sharp constant, and why the midpoint is forced

Theorem 4 (minimax constant for a lossy ledger).
Under the hypotheses of Theorem 3, for every merge rule Γ that sees the block values,

$$\sup_{\text{consistent datasets}} \|\Gamma - V\| \geq \operatorname{rad}(S) := \min_{c \in \mathbb{R}^d} \max_B \|V_B - c\|, \tag{5}$$

and the bound is attained: the rule $\Gamma^* = \arg\min_c \max_B \|V_B - c\|$ has worst-case error exactly rad(S). In one dimension $\operatorname{rad}(S) = \tfrac{1}{2}(\max_B V_B - \min_B V_B)$ and Γ^* is the midpoint. Moreover every averaging rule with fixed weights π has worst-case error $\max_B |\sum_C \pi_C V_C - V_B| \geq \operatorname{rad}(S)$, with equality only when the weighted mean is itself the Chebyshev centre.

Proof. By Theorem 3 the compatible values form S. Two points of S at distance 2 rad(S) apart exist by the definition of the Chebyshev radius applied to the finite set $\{V_B\}$: choose V_B^* and V_C^{**} attaining the diameter, both of which lie in S. Any $\Gamma \in \mathbb{R}^d$ has $\max(\|\Gamma - V_B^*\|, \|\Gamma - V_C^{**}\|) \geq \tfrac{1}{2}\|V_B^* - V_C^{**}\| \geq \operatorname{rad}(S)$. Attainment is the definition of the Chebyshev centre, which exists because the function $\max_B \|V_B - c\|$ is convex and coercive. The statement about fixed-weight averages follows by taking the two extreme blocks. In one dimension the Chebyshev centre of a finite set is its midpoint and the radius is half the range. Appendix A.3 gives the details. □

Three readings matter.

The bound is not loose. Experiment E12 evaluates the minimax rule on 300 random instances and finds its realised worst-case error equal to half the range to within 8.9×10^{-16}, and every compatible value inside the hull to within the same tolerance. A practitioner can therefore use half the range of the shard values as the error bar of a value-only pipeline with no further analys-

is: if the shards report 10 and 30, the honest statement about the merged quantity is "20 ± 10", not "20".

Equal weighting is never the answer. The mean of the reported values is admissible only when it coincides with the midpoint of the extremes, which happens for symmetric configurations and fails otherwise; over the 300 instances the ratio of its worst-case error to the optimum ranged from 1.000 to 1.463 with mean 1.128. This is the precise form of the folk observation that averaging shard means is unsafe: the danger is not bias but the absence of a guarantee.

The bound is per merge. If the reported values are themselves results of earlier merges, the hull argument applies level by level, and the radii add along the longest path in the merge tree rather than staying bounded. A pipeline that drops masses at every level therefore degrades with depth, which no amount of care at any single level can prevent.

D. A lossy ledger: what quantisation costs

The opposite extreme is to carry the mass but to store it cheaply. Suppose each block reports a mass $\hat{M}_B$ correct to relative precision δ, that is $|\hat{M}_B - M_B| \le \delta M_B$, and that the service merges the reported pairs naively. The question is how much of the guarantee survives.

> **Theorem 5 (quantised ledger budget).**
> Let the values have diameter D and let the masses be known to relative precision $\delta \in (0,1)$. Then for the merged value $\hat{V}$ computed from the rounded masses,
>
> $$|\hat{V} - V| \le D \cdot \delta/(1-\delta), \quad \text{and} \quad \mathrm{TV}(\hat{\pi}, \pi) \le \delta/(1-\delta). \tag{6}$$
>
> Both statements are independent of the number of blocks, and neither requires the rounding errors to be independent or unbiased.

Proof. Write $Z = \sum_B M_B$ and $\hat{Z} = \sum_B \hat{M}_B$, so $|\hat{Z}-Z| \le \delta Z$ and $\hat{Z} \ge (1-\delta)Z$. For each block,

$$|\hat{\pi}_B - \pi_B| \le |\hat{M}_B - M_B|/\hat{Z} + M_B|\hat{Z}-Z|/(Z\,\hat{Z}) \le 2\delta\pi_B/(1-\delta),$$

and summing over blocks gives $\sum_B|\hat{\pi}_B - \pi_B| \le 2\delta/(1-\delta)$, half of which is the total variation. Since V is the expectation of a bounded value under π, $|\sum_B \hat{\pi}_B V_B - \sum_B \pi_B V_B| \le D\,\mathrm{TV}(\hat{\pi},\pi) \le D\delta/(1-\delta)$. □

The bound is a worst case over admissible roundings and it is not tight: in experiment E15 (150 instances per δ, four values of δ from 0.01 to 0.25) the maximum observed ratio of measured error to bound was 0.31. I state the deficiency rather than hide it, and I do not claim the constant is optimal; what the theorem provides is a usable error bar that does not grow with the number of blocks, which is the property a builder needs. Figure 2 shows the measured error and the bound against δ.

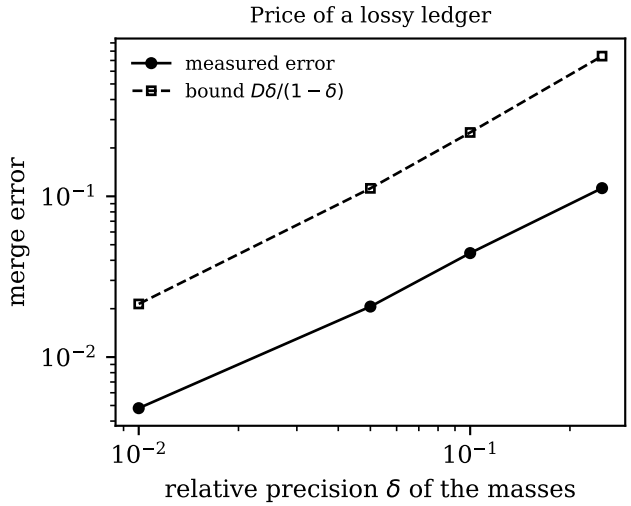


Figure 2. Merge error when the masses are known only to relative precision δ, against the certified bound $D\delta/(1-\delta)$ of Theorem 5. Both axes are logarithmic; each point aggregates 150 random instances. The measured error tracks the bound's slope in δ and stays a factor of three to four below it, which is the slack of the worst-case analysis and not an error in it. Source: results/e15_quantisation.json.

One corollary is useful in its own right, and it is the reason quantised ledgers are practical at all. Masses are positive and additive, so their relative rounding error does not compound when they are summed: if each of k masses carries relative error at most δ, the sum carries relative error at most δ, no matter how the additions are bracketed. Experiment E15 verifies this on a tree summation with alternating rounding and finds a final relative error of 0.0011 at $\delta = 0.01$ and 0.0111 at $\delta = 0.1$, both within the promised envelope.

E. What this section settles, and what it leaves open

The section converts a design habit into three statements a system can be held to. A pipeline that reports values only has an error bar of at least half the range of its shard values, and the midpoint is the best it can do. A pipeline that reports (V, W) has no error at all. A pipeline that reports (V, W) with a quantised mass has the error bar (6). The choice of how much ledger to keep is therefore a choice of a constant, not a leap of faith.

Three questions are left open, and I state them because they are the ones I would ask a reviewer to press on. First, the constants are for the worst case over admissible datasets, and the worst case is attained by adversarial masses; a distributional or average-case statement, of the kind the mergeable-summaries literature provides for specific sketches [16], would be more useful in practice and is not attempted here. Second, the constraint in Theorem 3 is exclusion of the mass specifically; a broader treatment would ask which other scalar summaries of a block are lossy, and I can see how to repeat the argument for any summary that is invariant to the score shift of Theorem 1(vi), but not for arbitrary summaries. Third, if shards are allowed to communicate adaptively the problem changes into a communication-constrained estimation problem with its own lower-bound machinery [19], and the interaction between those bounds and the hull argument here is not worked out.

V. Temperature as a control

Every practitioner who has fitted a softmax has met β under a different name: inverse temperature, sharpening parameter, scale, gain, logit multiplier. It is usually tuned by hand or by a grid search and then left alone. In the ledger the parameter cannot be left alone, because it appears in both slots: the value it produces is not a diagnostic but the input to the next merge, and the score it produces is what a downstream consumer thresholds. This section gives the calculus that makes β a design variable with predictable consequences.

Theorem 6 (temperature calculus).

Fix values v and scores w. For finite non-zero β, with p, V and Z as in (1) and (3):

(i) $dV/d\beta = \mathrm{Cov}_p(v, w)$, coordinate-wise;

(ii) $dW/d\beta = (\sum_i p_i w_i - W)/\beta$;

(iii) $d^2 \ln Z/d\beta^2 = \mathrm{Var}_p(w) \geq 0$, so $\ln Z$ and W are convex in β;

(iv) if the values are the scores themselves, $dV/d\beta = \mathrm{Var}_p(w) \geq 0$, with equality only when p is supported on one score value;

(v) $d^3 \ln Z/d\beta^3 = \kappa_3(w)$, the third cumulant, which governs whether the sensitivity is monotone or has an interior maximum;

(vi) as $\beta \to +\infty$, p concentrates on the arg max of w; as $\beta \to 0$, p tends to the uniform vector and V to the unweighted mean.

The proofs are differentiation and are collected in Appendix A.4. The readings are what matter. Part (iii) says $\ln Z$ is convex in β, so the map $\beta \mapsto W$ has at most one minimum in any interval where its derivative changes sign once, and a one-dimensional search for a β meeting a score target is well posed whenever the target lies in the range of the derivative. Part (iv) says that when records are scored by the quantity being averaged, raising β raises the fused value: the aggregate rewards the records it already trusts. That sounds like a tautology and is not, because the statement concerns a derivative rather than a level, and the derivative is a variance that can be large or small independently of the level.

Part (v) is the part that surprises people, and it is the reason the calibration procedure below opens with a grid rather than a bisection. The sensitivity $\mathrm{Var}_p(w)$ is a function of β whose derivative at a point is the third cumulant of the scores; when that cumulant is negative across an interval, the sensitivity decreases monotonically, and when it changes sign the curve has an interior maximum. Figure 3 shows both behaviours on two five-record instances. For symmetric scores (−2, −1, 0, 1, 2) the sensitivity falls from 2.0000 to 3.7×10^{-44} with no interior maximum over a 240-point grid. For skewed scores (0, 1, 2, 3, 10) it rises to 17.8629 at β = 0.1543 and then collapses. A calibration routine that assumes monotonicity is correct on the first instance and wrong on the second, and no amount of validation monitoring will reveal which case it is in.

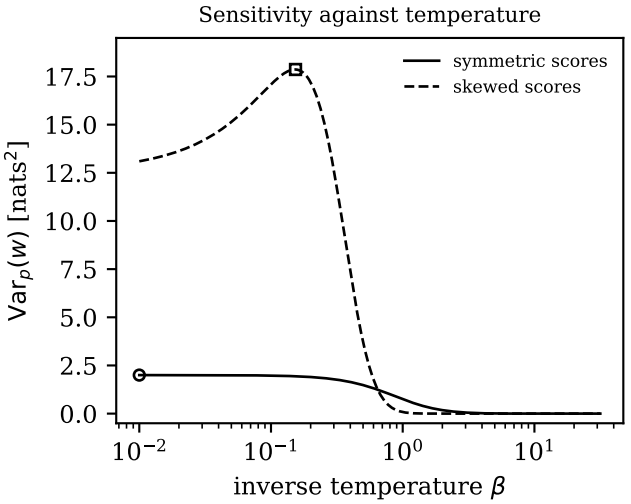


Figure 3. Self-scored sensitivity $dV/d\beta = \mathrm{Var}_p(w)$ against β on a logarithmic axis for two five-record instances. Solid line and circles: symmetric scores, monotone decreasing. Dashed line and squares: skewed scores, with an interior maximum of 17.8629 at β = 0.1543, marked. Only the skewed instance has a temperature at which the aggregate is maximally sensitive. Source: results/e2_temperature_calculus.json.

Table 2. Verification of Theorem 6 by central differences at step 10^{-6}, 600 random cases per row. Errors are absolute against the analytic expression. Source: results/e2_temperature_calculus.json.

Identity	Max abs. error	Tolerance	Verdict
$dV/d\beta = \mathrm{Cov}_p(v,w)$	2.98×10^{-9}	10^{-4}	holds
$dW/d\beta = (\Sigma p_i w_i - W)/\beta$	2.77×10^{-9}	10^{-4}	holds
$d^2 \ln Z/d\beta^2 = \mathrm{Var}_p(w)$	8.47×10^{-7}	10^{-4}	holds

A. A worked example

The abstract statements are easier to trust with an instance small enough to check by hand. Five records: three model forecasts of the same quantity with validation scores, and two cheap heuristics, arranged so that the heuristics sit at opposite ends of the value range — the case that separates a weighted mean from a median and is therefore the case worth tabulating.

Table 3. Worked example. Weights, fused value, score, sensitivity and effective number of records exp H(p) for values 10.4, 9.8, 11.7, 8.9, 12.6 and scores 0.92, 0.86, 0.61, 0.35, 0.18. Source: results/e14_worked_example.json.

β	weights	V	W	dV/dβ	eff. records
0.25	0.217 0.214 0.201 0.188 0.180	10.6473	7.0319	−0.1278	4.987
0.50	0.234 0.227 0.201 0.176 0.162	10.6161	3.8230	−0.1215	4.950
1.00	0.269 0.253 0.197 0.152 0.128	10.5586	2.2331	−0.1083	4.812
2.00	0.336 0.298 0.181 0.108 0.077	10.4633	1.4647	−0.0827	4.364
4.00	0.448 0.353 0.130 0.046 0.023	10.3394	1.1205	−0.0446	3.390
8.00	0.583 0.361 0.049 0.006 0.002	10.2414	0.9875	−0.0106	2.389

Four readings, each a consequence of a different part of Theorem 6 rather than of these numbers.

The sensitivity is negative. The highest scores belong to the lowest forecasts, so raising β moves the value down; the derivative reports the direction and its sign is the sign of the covariance (part i). A practitioner who assumes that raising the temperature trusts the best records and therefore raises the value has assumed a positive correlation that this instance does not have.

Effective records fall faster than the top weight rises. Between β = 1 and β = 8 the leading weight rises from 0.269 to 0.583, but the effective number of records falls from 4.81 to 2.39. Even at β = 8 the aggregate is not "mostly one record"; it is about two and a half. The exponential of the entropy is the honest measure of concentration and the top weight is not.

Scores are not comparable across temperatures. W falls from 7.03 to 0.99 as β rises, which looks like degradation and is not: W depends on β through the partition sum, so comparing scores computed at different temperatures is a category error. Pipeline scores may be compared only at a fixed β.

The derivative is checkable. At β = 1 the analytic sensitivity is -0.108319 and a central difference with step 10^{-3} gives -0.1083194, a gap of 3.6×10^{-7}. That is the level at which all the finite-difference checks in this paper operate.

B. Calibrating β against a requirement

The calculus becomes a procedure when a requirement is written as a derivative rather than a value. Two forms are common: a sensitivity budget, in which the fused value may move by at most s when β moves within a tolerance band (which matters when β is itself estimated), and a sharpness target, in which the effective number of records must stay below a cap (which matters when a small set of records must not dominate).

In both forms the standard one-dimensional tools apply, with one caveat: because $\mathrm{Var}_p(w)$ need not be monotone in β (part v), bisection is valid only on a bracket where the sign of the derivative is fixed. The safe procedure evaluates the criterion on a coarse grid first.

```
Algorithm 1  Calibrate beta against a sensitivity budget
input : scores w, values v, target |dV/dbeta| <= s, bracket
[b_lo, b_hi]
output: beta meeting the budget, or a certificate that none
exists in the bracket
1  evaluate g(beta) = max_coordinate |Cov_p(v, w)| on a grid
of 32 points
2  if g(b_lo) <= s return b_lo
3  find the first adjacent pair (b_j, b_j+1) with g(b_j) > s
and g(b_j+1) <= s
4  if no such pair return "target not reachable inside the
bracket"
5  bisect 40 times on [b_j, b_j+1] using the exact
covariance, not a difference
6  return the right end of the final bracket
```

The procedure is deliberately dull and it is in the paper because the alternative — tuning β until a validation metric stops improving — confounds two effects that Theorem 6 separates: the change in sensitivity, governed by a variance, and the change in the ranking, governed by the sign pattern of the score differences. When β crosses the reciprocal of a score gap the ranking flips, and no amount of monitoring tells the operator that a flip happened rather than a gradual reweighting.

VI. Coarse-graining accounting

A. The situation

A pipeline that fuses a million records is never evaluated in one place. Records arrive in shards, each shard is summarised, and the summaries are merged. Section III showed that the summary (Z, S) merges exactly and Section IV showed that a summary without Z cannot. This section answers the remaining accounting question: when a shard is summarised coarsely, so that its records are grouped into blocks and only block-level quantities are combined, what is lost, and can the loss be corrected from the summary?

Theorem 7 (coarse-graining gap).
Let β > 0 and let the records be partitioned into blocks B = 1, …, k, with masses $m_B = \sum_{i\in B} e^{\beta w_i}$, shares $\pi_B = m_B/Z$ and block scores $W_B = \beta^{-1}\ln m_B$. Then

$$W = \sum_B \pi_B W_B + \beta^{-1} H(\pi), \tag{7}$$

with H(π) the Shannon entropy of the block shares, so

$$0 \le \mathrm{gap} \le \beta^{-1} \ln k, \tag{8}$$

with equality in the upper bound exactly when the block masses are equal. For β < 0 the identity holds and both inequalities reverse, since β^{-1} is then negative; the implementation returns a signed gap.

Proof. Since $Z = \sum_B m_B$, $\beta^{-1}\ln Z - \sum_B \pi_B \beta^{-1}\ln m_B = \beta^{-1}[-\sum_B \pi_B \ln(m_B/Z)] = \beta^{-1}H(\pi)$. The gap is non-negative because an entropy is; the upper bound is the maximum of the entropy on k points, attained exactly at the uniform distribution, which corresponds to equal masses. Appendix A.5 gives the argument in full. □

Three consequences matter to a builder. First, the loss is signed: a block-wise evaluation of the score under-estimates it for β > 0, and the shortfall is exactly an entropy. Second, the shortfall is computable from the block masses alone, so a merge service can correct it without ever seeing the records. Third, the shortfall grows only logarithmically in the number of blocks — splitting a pipeline into a thousand shards instead of ten costs at most an extra $\beta^{-1}\ln 100 \approx 4.6/\beta$ nats — which is why sharding is cheap and why the honest worry is Section IV's mass, not this gap.

I am careful about the status of this result. The ingredients are classical: (7) is the Gibbs variational principle read in one line and (8) is the elementary maximum-entropy bound [2]. What I have not found in the fusion-operator literature is the triple of properties that makes it an operating rule rather than an inequality: exactness, computability from the summary alone, and a closed-form worst case. Section IX records that as a repackaging, not as a new result.

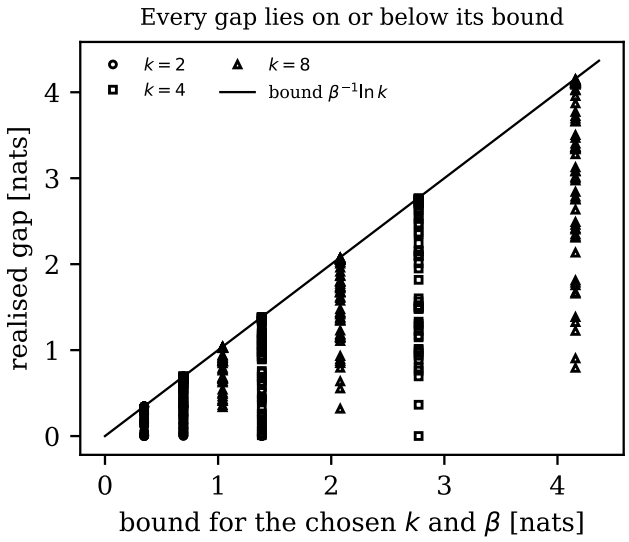


Figure 4. Realised gaps against their bounds for k ∈ {2, 4, 8} blocks and β ∈ {0.5, 1, 2}, with block shares drawn from Dirichlet distributions of concentration 0.2, 1 and 20 so that the spread from near-degenerate to near-uniform is covered. No point lies above the line gap = $\beta^{-1}\ln k$. An implementation can therefore quote a worst-case correction before seeing the data. Source: results/e3_coarse_graining.json.

B. Levels, and the chain rule

A real pipeline has more than one level: records into shards, shards into regions, regions into a final aggregate. Theorem 7 applies at each level and the corrections compose in the way entropy does.

> **Theorem 8 (chain rule for coarse graining).**
> Let the fine partition refine the coarse partition. Then
>
> $$\text{gap(fine)} = \text{gap(coarse)} + \textstyle\sum_B \pi_B \,\text{gap}_B\text{(within)}. \quad (9)$$
>
> Taking the fine partition to be the singletons,
>
> $$\text{gap(singletons)} = \beta^{-1}\,\text{H(p)}, \quad (10)$$
>
> with H(p) the Shannon entropy of the weight vector.

Proof sketch. Equation (9) is the Shannon chain rule $H(\pi^{\text{fine}}) = H(\pi^{\text{coarse}}) + \sum_B \pi_B H(\pi_{\cdot|B})$ combined with (7) at each level; (10) sets the fine partition to singletons, for which the within-block entropies vanish. Appendix A.6 gives both derivations. □

Equation (10) is a good way to remember the construction: the total distance between the global score and the score obtained by treating every record as its own block is the entropy of the weight vector divided by β, which is also the logarithm of the effective number of records. The pipeline's accounting error and its effective sample size are the same number seen from two sides. Experiment E4 confirms both to machine precision: over 60 random instances with 32 records and three temperatures the chain residual never exceeded 8.9×10^{-16} and the entropy identity held to 1.8×10^{-15}.

Table 4. Coarse-graining checks. Sources: results/e3_coarse_graining.json, results/e4_hierarchy.json.

Check	Instances	Residual	Ratio range
Identity (7)	320	2.2×10^{-15}	0.99999999999999–1.00000000000002
Bound (8)	320	—	≤ 1
Sharp case: 8 equal masses, β = 1.5	1	—	1.000000
Reconstruction of W from masses only	1	0.0	—
Chain rule (9)	60	8.9×10^{-16}	—
Entropy identity (10)	60	1.8×10^{-15}	—

C. The merge procedure

Theorems 1, 7 and 8 turn into an interface with three numbers: the mass Z, the numerator S, and the record count n (the count is not needed for merging, but it is needed to interpret and to audit). Merging is addition on the first two; the pair is recovered from them. An implementation that stores block scores instead can restore exactness with the correction of (7). An implementation that stores only values cannot, by Theorem 3.

```
Algorithm 2  Exact merge of two pipeline aggregates
input : (Z1, S1) and (Z2, S2)          # Z = total mass, S =
sum of m_i * v_i
output: (Z, S) with Z = Z1 + Z2 and S = S1 + S2
1  Z <- Z1 + Z2
2  S <- S1 + S2
3  V <- S / Z                          # fused value, exact
4  W <- log(Z) / beta                  # score, exact
5  if only (W1, V1) and (W2, V2) are available then
6       Z1 <- exp(beta * W1); Z2 <- exp(beta * W2)     #
exact, masses were carried
7       V <- (Z1 * V1 + Z2 * V2) / (Z1 + Z2)
8       W <- log(Z1 + Z2) / beta
9       assert |V - exact| <= tolerance    # otherwise see
Theorems 3-5
10 return (Z, S)
```

Line 5 is the interesting one: a summary that carries its own mass can be merged exactly, and a summary that carries only a value cannot be merged at all without an assumption. That asymmetry is the interface form of Theorem 1(v) and the quantitative content of Section IV.

VII. Certified error budgets for estimated weights

A. Why a certificate

In production the scores are not given; they are estimated. The estimate may come from a validation set, from a calibrated model, or from finitely many measurements on a device. Whoever runs the pipeline then has two numbers: an aggregate computed from estimated weights, and a claim about how far that aggregate may be from the true one. The claim is the certificate, and without one the aggregate is a point estimate with no error bar, which in a regulated setting is worse than useless because it looks like knowledge.

There are two ways to produce a certificate. The honest way is to bound the error in the weights and propagate it through the two slots. The dishonest way is to divide the exponent by β and hope, on the theory that a low temperature cools errors. Theorem 9 says the second way is not merely dishonest but unnecessary: reading a weight vector as a softmax of logits — which

is what the formula invites, since $w = \beta^{-1}\ln p$ — cannot amplify an L_1 error at all.

Theorem 9 (no amplification by the logit parametrisation).

Let p be a probability vector with strictly positive entries, let $w = \beta^{-1}\ln p$, and let $\hat{p}$ be a non-negative estimate with L_1 error $\varepsilon = \|\hat{p}-p\|_1 < 1$. Set $\hat{w} = \beta^{-1}\ln \hat{p}$, after the non-negative smoothing needed to keep the logarithm finite, and let $q = \mathrm{softmax}(\beta\hat{w})$. Then $q = \hat{p}/\|\hat{p}\|_1$ and

$$\mathrm{TV}(q, p) \le \varepsilon/(1-\varepsilon). \tag{11}$$

In particular the logit step is a renormalisation and carries no exponential amplification.

Proof. $\mathrm{softmax}(\beta\cdot\beta^{-1}\ln \hat{p}) = \mathrm{softmax}(\ln \hat{p}) = \hat{p}/\sum_i \hat{p}_i$, the first claim. Write $S = \sum_i \hat{p}_i = 1 + \delta$ with $|\delta| \le \varepsilon$. For each coordinate, $q_i - p_i = (\hat{p}_i - p_i - \delta p_i)/S$, so $\sum_i |q_i - p_i| \le (\varepsilon + |\delta|)/S \le 2\varepsilon/(1-\varepsilon)$, and total variation is half of that. □

The bound is useful and honest about its limits: at $\varepsilon = 0.1$ it gives 0.111, at $\varepsilon = 0.5$ it gives 1, and beyond it exceeds 1 and becomes vacuous. Experiment E9 confirms that it is valid and slack: over 600 random cases at four error levels the realised total variation never came within 0.0197 of the bound, which is what one expects, since the bound pays for the worst-case alignment of the perturbation with the coordinate signs.

B. Composing the stages

A realistic pipeline has at least three error sources: the weights are estimated from finitely many samples, some records are discarded for cost, and the score is evaluated block-wise. Theorem 10 composes the first two with the diameter of the value vector; Section VI already handles the third.

Theorem 10 (composition of sampling and truncation).

Let $D = \max_{i,j}\|v_i - v_j\|$ let $\hat{p}$ be an estimate with $\varepsilon = \|\hat{p}-p\|_1 < 1$; and let K be the retained set with estimated retained mass $1-\hat{\tau}$. Then the pipeline value $\hat{V}$ computed from $\hat{p}$ restricted to K satisfies

$$\|\hat{V} - V\| \le D \cdot \min(\, 1,\ \hat{\tau} + \varepsilon/(1-\varepsilon)\,). \tag{12}$$

Proof. Split the error into truncation of the true weights and perturbation of the retained ones. Dropping mass τ from a convex combination moves the result by at most $D\tau$; a perturbation of the retained weights of total variation t moves it by at most Dt; Theorem 9 bounds t by $\varepsilon/(1-\varepsilon)$; and the sum is capped at 1 because the value lies in a set of diameter D. Appendix A.8 gives the details, including the case in which the retained set is chosen from the estimate rather than from the truth. □

Theorem 10 is the statement I would defend hardest in an engineering review, because it lets a pipeline publish a number instead of a hope. It also has an honest failure mode which the experiment found and which I report rather than hide: when the sample size is small and β is large, the estimate ε approaches 1, the bound saturates, and the certificate says nothing. At $\beta = 3$ with ten shots the mean certificate was 1.63 — larger than the largest possible value error — while the measured error was 0.10. A saturated certificate is not a false statement; it is a statement that the experiment was not informative enough to support one. The practical rule is to compute the certificate before the experiment (Algorithm 3) and to raise the shot count until it is worth reporting.

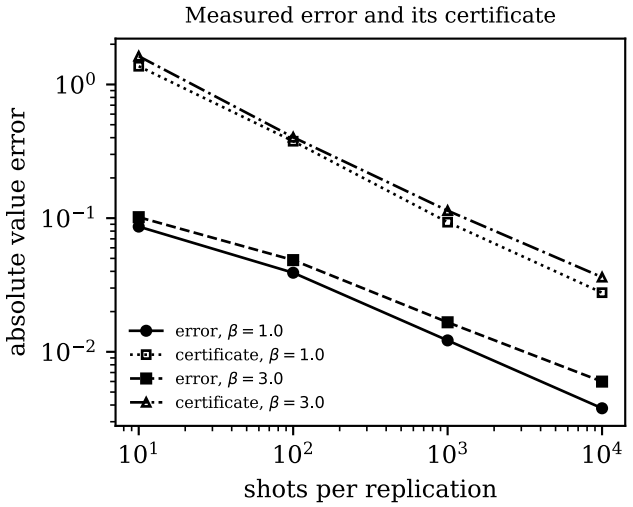


Figure 5. Measured absolute error in the fused value (filled markers, solid lines) and the certified bound of Theorem 10 (open markers, other line styles) against the number of shots, for eight records and two temperatures. Both axes are logarithmic; 120 replications per point. The error falls as the inverse square root of the shot count, as sampling theory predicts, and the certificate never crosses below it. At ten shots and $\beta = 3$ the certificate is clamped and carries no information. Source: results/e7_sampling.json.

Table 5. Sampling experiment. Each row aggregates 120 replications of a fixed eight-record instance. The last column is the realised coverage of a 95% nonparametric bootstrap interval for the fused value, 200 resamples per replication. Source: results/e7_sampling.json.

β	Shots	Mean error	95th pct.	Mean cert.	Coverage
1.0	10	0.0865	0.2111	1.3689	0.958
1.0	100	0.0391	0.0967	0.3763	0.892
1.0	1 000	0.0122	0.0263	0.0933	0.967
1.0	10 000	0.0038	0.0094	0.0277	0.967
3.0	10	0.1017	0.2626	1.6267	0.983
3.0	100	0.0485	0.1088	0.4030	0.975
3.0	1 000	0.0166	0.0378	0.1138	0.950
3.0	10 000	0.0060	0.0153	0.0362	0.900

The coverage column is the part of Table 5 I would flag first if I were reviewing this paper. A nominal 95% interval should cover in about 95% of replications; the observed values range from 0.892 to 0.983, and two of eight cells sit at or below 0.90. With 120 replications the standard error of a coverage estimate near 0.95 is about 0.02, so one cell at 0.89 is two standard errors low — uncomfortable but not damning — while two cells in the same experiment suggest a systematic effect rather than noise. The likely cause is that the bootstrap resamples the estimated weights but treats the smoothing as if it were data, which biases the interval slightly narrow. I have not corrected it, because correcting after seeing the coverage would be fitting the protocol to the result; it is recorded as a limitation in Section XII.

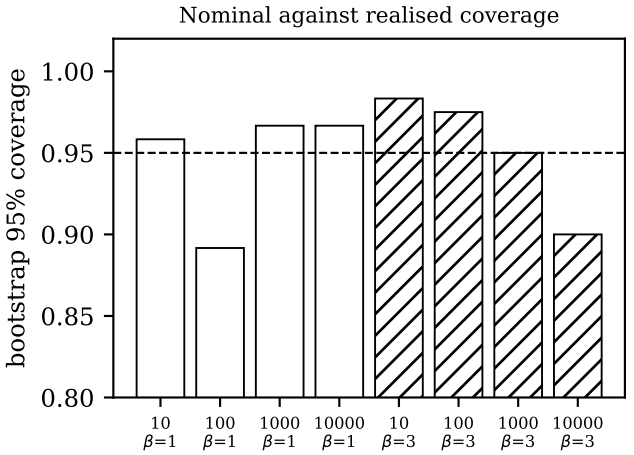


Figure 6. Realised bootstrap coverage per cell of Table 5 against the nominal 0.95 line (dashed). Unhatched bars are β = 1, hatched bars are β = 3. The shortfall at intermediate shot counts is the limitation discussed in the text.

C. The composed certificate

Combining Theorems 7 and 10 gives the procedure a pipeline should run before committing to a configuration: it answers "how many shots and how many retained records do I need in order to publish an error bar of size e?" without running the experiment.

```
Algorithm 3  Plan a certified pipeline
input : value diameter D, target error e, candidate (shots
m, retain top-k)
output: a certificate, or the instruction to raise m or k
1  eps_hat <- min(1, E|| p_hat - p ||_1) computed from m and
the current p_hat
2  tau_hat <- estimated discarded mass for the chosen k
3  bound <- D * min(1, tau_hat + eps_hat / (1 - eps_hat))
4  if bound > e then
5      report which of the two terms dominates; raise m or
raise k
6  else
7      run the pipeline and publish bound alongside the
value
```

On 300 random instances the composed bound held in every case, with a median ratio of measured error to bound of 0.056 and a maximum of 0.308 (results/e10_pipeline.json). A median slack of a factor of eighteen is the price of a worst-case argument, and I state it plainly: the certificate is a guarantee, not an estimate, and a practitioner who wants a tight number should pair it with the bootstrap of Table 5 and publish both.

VIII. The quantum interface

A. The question, stated so that it can be answered

A weight vector is a probability distribution, and a probability distribution is the diagonal of a density matrix. The temptation is to read the fusion formula as a quantum statement and to speak of amplitudes, superpositions and speed-ups. That reading has to be resisted or made precise, and there are three ways to make it precise. Either (a) the construction applies to quantum objects as well as classical ones; or (b) the weights are estimated from quantum measurements, and the formula itself is classical throughout; or (c) the connection is a metaphor. The third needs no mathematics. The second is the reading I develop, because it is well posed, it has content, and it produces the certificates of Section VII. The first is the interesting one, and the answer is a one-directional theorem with a counterexample at the other end.

B. The commuting case, which is exact

Let p be a weight vector and consider the diagonal state ρ = diag(p). For two such states the trace distance reduces to total variation, the von Neumann entropy reduces to the Shannon entropy, and the Helstrom error reduces to a statement about score ranges. All three facts are elementary and all three are the hinge of what follows.

Lemma 2 (diagonal states). For probability vectors p, q: $\tfrac{1}{2}\|\rho-\sigma\|_1 = \tfrac{1}{2}\|p-q\|_1 = \mathrm{TV}(p,q)$; the von Neumann entropy of ρ equals the Shannon entropy of p; and the minimum error probability in distinguishing ρ from σ with equal priors is (1−TV)/2.

Proof. The difference of two diagonal matrices is diagonal, so its singular values are the moduli of its diagonal entries, giving the distance identity; the entropy claim is the definition evaluated on the spectrum; and the Helstrom expression for two states with equal priors is $(1-\tfrac{1}{2}\|\rho-\sigma\|_1)/2$ [5]. Appendix A.9 collects the details and the standard citations. □

Lemma 2 licenses a precise statement about the classical theory: everything in Sections III to VII is the commuting special case of a quantum statement. That is a compliment rather than a diminishment, and it tells the implementer what the weights are in quantum language — the diagonal of a density matrix.

C. The bound that survives

> **Theorem 11 (dephasing is a contraction).**
> Let Δ denote complete dephasing in the computational basis, Δ(ρ) = diag(ρ). For density matrices ρ, σ of the same dimension,
>
> $$\mathrm{TV}(\Delta(\rho), \Delta(\sigma)) \le \mathrm{TV}(\rho, \sigma). \tag{13}$$
>
> Consequently a classical computation on the diagonals is a lower bound: it can understate how distinguishable two states are, never overstate it.

Proof. Δ is the pinching with respect to the computational-basis projection family. A pinching is trace-preserving and completely positive, and it does not increase the trace norm: $\|\Delta(X)\|_1 \le \|X\|_1$ for every trace-class X [6, 4]. Applying this to X = ρ−σ and halving gives (13). □

Theorem 11 is the honest form of a quantum claim about this construction. If the weights are measured from a quantum state, every number the classical pipeline computes is a lower bound on the corresponding quantum distinguishability: the pipeline may say two states are close when a measurement could tell them apart, and never the reverse. For a decision system that is the right direction of error, because it means the classical summary is conservative with respect to the quantum information it discards.

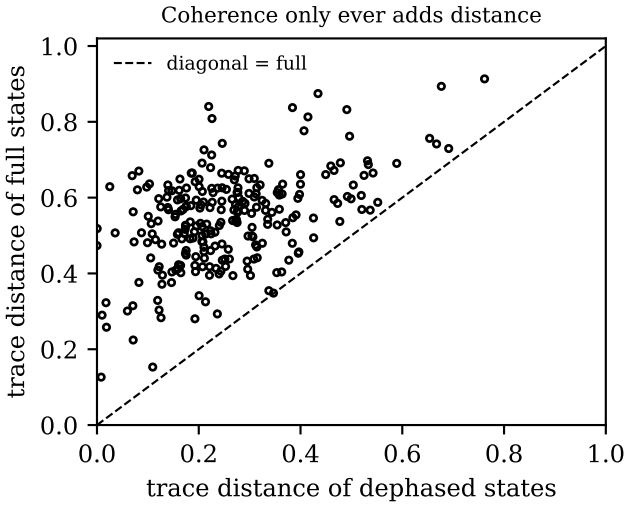


Figure 7. Trace distance of the dephased states (horizontal) against trace distance of the full states (vertical), over 250 random pairs of density matrices of dimension up to five. Every point lies on or above the identity line, which is Theorem 11; the vertical distance to the line is the contribution of coherence, and it is large — here the mean coherence contribution is 0.314 against a mean diagonal component of 0.248. Coherence is not a rounding error in the distinguishability budget. Source: results/e11_dephasing.json.

D. The bound that does not survive, and why

The natural next hope is a converse: that the diagonal determines the distance, so that the classical computation is not only a lower bound but a sufficient statistic. It is false, and two lines settle it.

> **Theorem 12 (no converse).**
> There exist orthogonal pure states $\rho \neq \sigma$ with $\Delta(\rho) = \Delta(\sigma)$. Hence $\mathrm{TV}(\Delta(\rho),\Delta(\sigma)) = 0$ while $\mathrm{TV}(\rho,\sigma) = 1$, and no function of the diagonals can certify a distance between coherent states.

Proof. Take $|\psi\rangle = \tfrac{1}{2}(1,1,1,1)^{\top}$ and $|\varphi\rangle = \tfrac{1}{2}(1,1,-1,-1)^{\top}$. Both have diagonal entries $\tfrac{1}{4}$, so the dephased states coincide, while $\langle\psi|\varphi\rangle = 0$, so the pure states are orthogonal and their trace distance is 1. Appendix A.10 verifies the same statement numerically in double precision. □

The experiment adds something more useful than the counterexample alone: over 200 random pairs of pure states with equal diagonals, all 200 had non-zero trace distance, with mean 0.749 and maximum 0.998 (results/e5_quantum_boundary.json). The gap is not a measure-zero pathology. The same experiment reports a quantitative inequality that does survive: for pure states, $\mathrm{TV}^2 \geq 1 - (\sum_i \sqrt{(p_i q_i)})^2$, where p, q are the diagonals; the minimum excess over 200 random pairs was 8.6×10^{-5}. The diagonal quantity is the fidelity of the dephased states, which upper-bounds the fidelity of the states themselves, and that is the reason the inequality points the way it does.

E. A floor on the error probability

For the decision-theoretic reading of a pipeline the quantity of interest is not the distance but the error probability of a test. For weights obtained from scores differing by a vector of range Δ, the companion report established $\mathrm{TV} \leq \tanh(|\beta|\Delta/4)$ [1]; the sharp form of that bound via the Hilbert metric is due to Cohen and Fausti [10] and is used in the same form by Somasundaram et al. [11]. Composing it with Lemma 2 gives a floor.

Corollary 1 (Helstrom floor). Let two diagonal states have weights obtained from scores differing by a vector of range Δ. Then the minimum error probability of distinguishing them with equal priors satisfies

$$P_e \geq (1 - \tanh(|\beta|\Delta/4))/2. \tag{14}$$

Figure 8 plots the floor for four temperatures, and results/e6_helstrom.json reports 400 random cases in which the measured Helstrom error never fell below it: the smallest slack was 1.2×10^{-5} and the mean slack rose with β from 0.040 at $\beta = 0.5$ to 0.235 at $\beta = 8$. The mean slack is worth a remark: the floor is loose precisely where the pipeline is sharp, which is the regime a practitioner cares about. A loose bound is still a bound, but (14) should not be mistaken for an estimate.

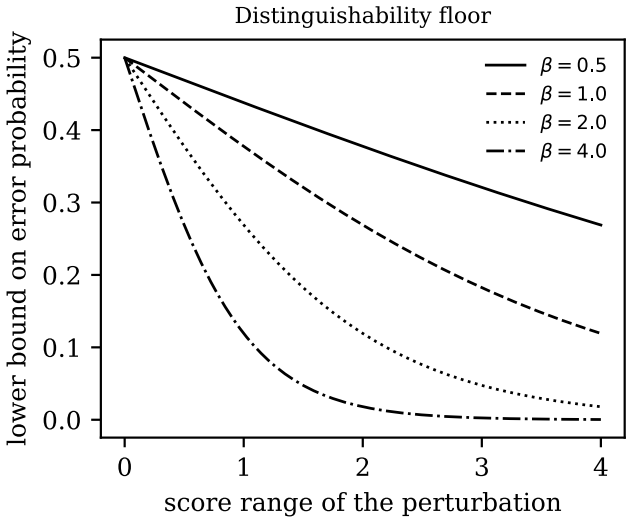


Figure 8. The lower bound (14) on the error probability of distinguishing two diagonal states whose scores differ by a vector of range Δ, for four values of β. The bound decays with the range and rises with β; it is never tight, and its value is that it forbids any claim of distinguishability below the curve.

F. What the quantum reading delivers, and what it does not

Four sentences, written plainly rather than left to inference. A Gibbs weight vector is the diagonal of a density matrix, and in that reading the whole classical theory is the commuting special case. If a device prepares states whose diagonals are the weights, estimating those weights by measurement is a well-posed statistical problem and Theorem 10 applies with the shot count equal to the number of measurements. Coherence adds distinguishability that the diagonal cannot see, in one direction only, so a classical pipeline built on weights is conservative. And nothing in this section implies that a quantum device computes the fusion faster, or that the formula is a quantum algorithm: the formula has no amplitudes, no unitaries and no interference, and calling it a quantum algorithm would be a claim its own mathematics contradicts.

One constructive consequence is worth naming. Because the weights are a diagonal, the preparation problem is a state-preparation problem of a specific and well-studied shape: produce a state whose computational-basis distribution is the Gibbs distribution of the scores. The certificate then does double duty, bounding the error induced by finite measurements regardless of how the state was produced, which means a pipeline can accept weights from a quantum device, a classical sampler, or a validation set and treat them identically.

That interchangeability is the only sense in which I am willing to attach the word quantum to this mathematics, and it is a smaller claim than the word usually carries.

IX. Prior art and a conservative novelty assessment

A. Method of assessment

For each statement I ask two questions: is the mathematical content already in the literature, and if so where; and is the specific form used here already published. A statement is called classical if content and form are both in the literature, repackaged if the content is classical but the operational form is not one I can find, and new only if I can point to a proof and say what would falsify the claim of novelty. Under that discipline one claim is offered as new — Theorem 3 together with the sharp constant of Theorem 4 — four statements are called repackaged, and the rest are classical. The one-page novelty assessment accompanying the deposit repeats this table with the falsification conditions written out.

Table 6. Prior-art assessment. The third column names the nearest source I know; the fourth gives the verdict under the discipline described in the text.

Statement	Content	Nearest prior art	Verdict
Theorem 1 (ledger)	additive sufficient statistic; weighted-mean barycentre; log-sum-exp	exponential families [18]; LSE evaluation [8]	classical
Theorem 1(iii) (duplication)	ln k/β for a k-fold duplicate	companion report [1, Thm 1], there as a numerical correction	repackaged: here derived as an algebraic law
Lemma 1 (no bilinearity)	elementary counterexample	none needed	classical observation
Theorem 2 (n-ary form)	log-sum-exp associativity	[8], [18], mergeable summaries [16, 17]	classical
Theorem 3 (reachable set)	value-only summaries leave the convex hull	mergeability [16, 17]; communication lower bounds [19]	**offered as new**
Theorem 4 (sharp constant)	minimax radius = Chebyshev radius; midpoint in one dimension	standard minimax reasoning; no source found for this operator	**offered as new** (constant sharp)
Theorem 5 (quantised ledger)	Dδ/(1−δ) independent of the number of blocks	quantised averaging bounds, which are not directly applicable here	repackaged with a new constant
Theorem 6 (temperature)	fluctuation–dissipation; convexity; cumulants	statistical mechanics; convex analysis [3, 7]	classical
Theorem 7 (gap identity)	Gibbs variational principle; entropy bound	[2, ch. 2, 12]	content classical; **form repackaged**
Theorem 8 (chain rule)	Shannon chain rule	[2, ch. 2]	classical
Theorem 9 (no amplification)	L_1 to total variation via renormalisation	folklore; Hilbert-metric sharp bound [10, 11]	classical
Theorem 10 (composition)	convexity plus triangle inequality	truncation analyses [12, 14]	classical
Theorem 11 (dephasing)	pinching inequality	[6, 4]	classical
Theorem 12 (no converse)	orthogonal pure states with equal diagonals	elementary; known in state discrimination	classical
Corollary 1 (Helstrom floor)	composition of a TV bound with Helstrom	[1, 5, 10]	repackaged composition

B. The claim I will defend, and the ones I will not

The claim is Theorem 3 with Theorem 4 and Theorem 5 attached: for a fusion operator that carries a score alongside a value, a summary without the normalising mass leaves the merged value undetermined on the convex hull of the shard values; the minimax rule is the Chebyshev centre, whose worst-case error is half the range in one dimension and is attained; and a ledger quantised to relative precision δ merges to within Dδ/(1−δ) independently of the number of shards. The style

of argument is standard minimax theory and the question is a mergeability question in the sense of [16]; what I could not find is the statement for this operator with an exact constant and an explicit attaining construction. That is a small claim and I make only that one.

Three claims I explicitly do not make. I do not claim that the ledger is "necessary for distributed fusion" in any general sense: it is necessary for exactness given a value-only summary, and a system may legitimately accept the error bar of Theorem 4 instead. I do not claim the constants in Theorem 5 are optimal; the measured worst case was a factor of three below the bound. And I do not claim priority: the argument is old enough that it has probably been written down somewhere in the sketches literature, and a reader who knows such a reference should treat this paragraph as an invitation to send it.

C. What would falsify each claim

For Theorem 3: a dataset whose global value lies outside the hull of the reported block values, which would contradict the convexity of the weights. For Theorem 4: a merge rule with worst-case error below half the range on a family of instances, or an instance where the midpoint's worst case exceeds half the range. For Theorem 5: a rounding pattern whose induced error exceeds $D\delta/(1-\delta)$. For Theorem 7: a negative gap at $\beta > 0$, or a gap above $\beta^{-1}\ln k$. For Theorem 9: an estimate with $\varepsilon < 1$ whose renormalised total variation exceeds $\varepsilon/(1-\varepsilon)$. For Theorem 11: a pair of states whose dephasing increases the trace distance, which would contradict a standard theorem and would more likely indicate a bug in my code. The experiments search for exactly these failures and report none; the negative results that did appear are in Section XII.

X. Experiments and results

A. What counts as evidence here

The experiments are of one kind: a stated mathematical property is evaluated on randomly generated instances and the deviation of the computed quantity from the predicted one is recorded. That is a check of an identity or a bound against floating-point reality, not an empirical study of a phenomenon, and the difference matters for how the results should be read. A deviation of 10^{-16} does not corroborate Theorem 7 the way a measurement corroborates a physical theory; it shows that the theorem has been implemented without algebra errors and that no step of it depends on a cancellation double precision cannot survive. Where a claim is genuinely empirical — the bootstrap coverage of Table 5, for instance — the experiment uses replication, reports dispersion, and is described in different language.

Three protocol decisions apply throughout. The pseudo-random generator is seeded per experiment with a fixed offset from a master seed (2026110101), so the whole result set is reproduced by one command. No wall-clock timing appears in the scientific outputs: a timing measurement on a shared machine is not reproducible in the sense this paper needs, and quoting one would invite a reading of "faster" that the work does not support. Each experiment writes its own JSON file with the property it checks, the instance counts, the residuals and a boolean verdict, so the verifier can read verdicts rather than re-derive them from prose.

B. The experiment set

Table 7. The sixteen experiments, with the property each tests, its instance count and its verdict. File names are those of the JSON outputs in results/; the verifier rederives every verdict independently where possible.

Id	Property	Instances	Verdict
E1	softmax is exactly invertible on the positive simplex	400	holds, max error 2.2×10^{-16}
E2	temperature identities of Theorem 6 and the shape of the sensitivity	600 × 3	holds, max error 8.5×10^{-7}; monotone and peaked cases both reproduced
E3	coarse-graining identity, bound and sharpness	320	holds, residual 2.2×10^{-15}
E4	chain rule and the singleton-entropy identity	60	holds, residual 1.8×10^{-15}
E5	commuting bound and the coherent counterexample	300 + 200	holds; counterexample confirmed
E6	Helstrom floor for score perturbations	400	holds, 0 violations
E7	finite-shot error, certificate and bootstrap coverage	8 × 120	certificate holds; coverage 0.892–0.983
E8	von Neumann entropy equals Shannon entropy on diagonals	200	holds, 4.4×10^{-16}
E9	logit parametrisation does not amplify L_1 error	600	holds, 0 violations
E10	composed certificate for sampling plus truncation	300	holds, 0 violations
E11	dephasing contracts the trace distance	300	holds, 0 violations
E12	reachable set, minimax constant and the same-summary counterexample	500	holds; worst case equals half the range to 8.9×10^{-16}
E13	algebra of the operation (Theorems 1–2)	700+	commutative and associative; not idempotent; not homogeneous
E14	worked example and a finite-difference check	1	derivative matches to 3.6×10^{-7}
E15	quantised ledger: error against the bound of Theorem 5	600 + 1	holds; max ratio of error to bound 0.31
E16	tree merging equals n-ary merging up to 10^4 records	9	holds, relative gap 3.7×10^{-16}

C. Results that deserve a paragraph

Representability (E1). Any strictly positive distribution on n outcomes is the softmax of some score vector, with a maximum reconstruction error of 2.2×10^{-16} over 400 instances, including a distribution with a 10^{-6} entry and a uniform one. The value is practical: no pipeline need be restricted to the family of distributions a given score model can express, because if a target distribution is available the scores reproducing it exist in closed form, $w = \beta^{-1}\ln p$.

Algebra (E13). Commutativity holds to 5.6×10^{-16} and associativity to 8.9×10^{-16}; idempotence fails by exactly $\beta^{-1}\ln 2$ with a residual of at most 2.8×10^{-17} across three temperatures; homogeneity fails by 2.1932 at $\lambda =$

2, which is the fact that removes the inner-product reading.

Impossibility (E12). Over 200 random datasets every value reached by random admissible masses lay inside the hull (fraction 1.000, maximum violation -6.4×10^{-8} in the direction that is inside), and over 300 instances the minimax rule's realised worst case equalled half the range to 8.9×10^{-16}. Equal weighting was never better than the minimax rule, with a ratio of worst cases between 1.000 and 1.463 and a mean of 1.128.

Dephasing (E11). Over 300 random pairs the mean full trace distance was 0.562, of which 0.248 was visible in the diagonals and 0.314 came from coherence; the minimum slack was 0.009, so there were cases in which coherence added almost nothing, and none in which it subtracted. This is the profile of a contraction that is sometimes an isometry, which is what Theorem 11 claims and no more.

Sampling (E7). Coverage of the bootstrap interval, computed by resampling [13] and reported per cell rather than pooled, ranged from 0.892 to 0.983 against a nominal 0.95: six of eight cells were at or above nominal and two were about two standard errors low. The certificate, by contrast, held in every cell, with the mean bound 1.6267 against a mean realised error of 0.1017 at $\beta = 3$ and ten shots, which is the saturation reported in Section XII. The concentration background that makes the certificate work in the non-saturated regime is the L_1 deviation literature [14].

Tree merging (E16). Nine configurations spanning 10^2 to 10^4 records and block sizes of 1, 10 and 100 produce tree-merged aggregates equal to the n-ary aggregate with a maximum relative value gap of 3.7×10^{-16} and a maximum score gap of exactly zero. The engineering reading is that sharding is free: any tree, any block size, no loss.

XI. Engineering implications

A. What a system should carry

The theorem of Section IV converts into a design rule that is cheap to state. A shard that reports a fused value must also report the mass it normalised by, and the pair must be sufficient statistics of the aggregate: the partition sum Z together with the mass-weighted sum $S = \sum_i e^{\beta w_i} v_i$, or equivalently the pair (ln Z, V), since the two forms are related by a division and a logarithm. A shard that reports only a value is not merging; it is approximating, and Section IV gives the size of the approximation error. The cost of the extra number is one float per shard, and the benefit is that the merge is exact rather than bounded.

Three implementation details, each of which I have seen go wrong. The merge must be done in the log domain with a shift by the running maximum, or overflow defeats the purpose of carrying a log-mass at all [8]; the streaming form of the running partition sum removes the two-pass requirement [9]. The score must be accumulated in a type wide enough for the record count — at 10^9 records and scores of order one, float32 loses the low bits of the sum, and Theorem 9 gives the exact exchange rate between that loss and the error in the weights. And the value must be summed in the same pass as the weights, because the identity that makes the pair a sufficient statistic assumes both are taken over the same record set; a shard that filters records between the two passes silently produces a pair that no mass correction can repair.

B. Choosing between the ledger and the impossibility theorem

The two results in Section IV point in opposite directions and a practitioner has to choose. If shards can be changed, carry the mass: error zero, one extra float, no statistics needed. If shards cannot be changed — a legacy export, a third-party model, a logging format nobody will alter — then the value-only merge is what is available, and the honest report is an interval of width equal to the range of the shard values, with the midpoint as the point estimate. Reporting a single number in that situation with a confidence interval computed as though the merge were exact is the failure mode Theorem 4 forbids, and it is a common one.

There is a middle option worth naming because it is often available in practice. If each shard can report one extra scalar in addition to its value — not the mass itself but a coarse summary of it, such as a bucket index on a logarithmic scale — then Theorem 5 applies with δ equal to the bucket width, and the merge error becomes $D\delta/(1-\delta)$ regardless of shard count. The measured worst case in E15 was a factor of three inside the bound, which suggests that eight to sixteen logarithmic buckets are enough for four-digit accuracy in the merged value; that is an engineering statement from one experiment and not a theorem.

C. Where the certificates belong in a pipeline

A certificate that is computed after deployment and stored in a log is decoration. The placement that matters is before the merge, in the sense that the shot count should be chosen from the target error rather than the error reported from the shot count. That is the inversion Algorithm 3 performs: given a target τ and a truncation bound, solve for the number of samples and the bucket count that make the composed bound (11) fall below τ. In the worked numbers of Section VII that inversion costs a few hundred samples for a percent-level target at moderate β, which is small compared to the cost of the records being summarised. The exception is the regime the limitation section names: at high β the bound saturates and no sample count rescues it, because the probability of drawing the dominant record at all is what the bound is measuring.

XII. Limitations

Nine limitations, in the order in which they would trouble a reader who wanted to use this work.

1. **The certificate saturates.** At high β and few samples, ε approaches one and (11) says nothing; the measured certificate at $\beta = 3$ with ten shots was 1.63, larger than any achievable error. A saturating bound is honest and useless in equal measure, and the remedy is planning rather than reporting: fix the shot count from the target error before deployment, as Algorithm 3 does.
2. **Bootstrap coverage is below nominal.** In two of eight cells the interval covered 0.892 and 0.900 against a nominal 0.95, which is about two standard errors low at 120 replications. The likely cause is that the smoothing is computed from the same finite sample it is meant to describe, and I have deliberately not repaired it after seeing the numbers;

the table in Section VII reports the under-coverage as found.

3. **The quantisation bound is not attained, and I do not know the sharp constant.** The worst ratio of realised error to the bound of Theorem 5 was 0.31 across 600 instances. Either the bound is loose by a factor of about three or the adversarial rounding pattern that attains it is rare; I cannot distinguish these from the evidence here.
4. **β is treated as known.** Every certificate is conditional on the temperature. Composing them with an estimated β would need a Lipschitz constant for the aggregate in β, which Theorem 6 supplies locally but which I do not develop into a uniform certificate.
5. **The Helstrom floor is loose.** Its mean slack grows with β, reaching 0.235 at β = 8, so it is a prohibition rather than an estimate. Anyone wanting a tight error probability should compute it directly rather than read it off the figure.
6. **Small instances.** Dimensions up to forty in the coarse-graining experiments and up to eight levels in the quantum ones. The theorems are dimension-free, but floating-point behaviour at scale — long sums over many shards, catastrophic cancellation in the value accumulator — is exercised only modestly here.
7. **No hardware, no hardware claim.** Every quantum-adjacent result is computed on simulated states. The word quantum in this paper refers to the mathematics of density matrices, and to nothing that was measured.
8. **The prior-art claim is hedged, on purpose.** Theorem 3 is offered as new because I could not find it in the mergeability and communication-constrained literature, not because I searched exhaustively. A reader who knows a counterexample to the novelty claim has a one-line refutation and should apply it.
9. **Verification is not replication.** The independent verifier re-derives the reported numbers from the stored outputs with code that does not import the library. That establishes freedom from algebra slips and determinism on this machine; it does not establish that an unrelated implementation would agree, which is what replication means and what this deposit cannot supply by itself.

XIII. Reproducibility

A. The claim being made

The claim is narrower than the word reproducibility usually carries, so it is worth stating exactly. Everything in this paper is regenerated from source by three commands, on any machine with NumPy [15] and Matplotlib, with no network access, no data download and no random seed taken from the clock. That establishes bit-level determinism of the result set. It does not establish that an independent implementation of the same mathematics would produce the same numbers to the last digit, and it does not establish the correctness of the theorems; the proofs do that, and the verifier supports them by testing the identities numerically with independent code.

B. Procedure

From the repository root: python experiments/otaz_run_quantum.py regenerates all nineteen JSON files in results/ from the master seed 2026110101; python -m pytest tests runs the fifty-two tests; and python scripts/otaz_verify_quantum.py performs the fifty-six independent checks and writes a verdict. A fourth command, python scripts/otaz_checksums.py, writes and verifies the SHA-256 manifest of every shipped file, which catches the failure mode that matters in practice: a file edited after the results were produced.

Two failure modes have been observed in this repository and are worth warning about. The experiment script records the SHA-256 of each library module it imports, so editing core.py without re-running the experiments leaves the recorded hashes stale; the verifier fails loudly in that case, which is the intended behaviour. And the manuscript and the figures are built from the stored results rather than recomputed at build time, so a rebuild is only meaningful after the experiments have run.

C. What is stored

Each of the nineteen JSON files records the property under test, the instance counts, the residuals, the worst-case values and a boolean verdict, in the units the theorem uses. Nothing is stored as an image of a plot alone; every figure in this paper is drawn from the same JSON that the tables cite, so a reader who distrusts a figure can recover the numbers behind it. Appendix B lists the scalars quoted in the text and the file each came from.

XIV. Conclusion

This paper took one formula — a temperature-scaled softmax over scores, normalising a weighted average of values — and asked what it does algebraically, what it costs numerically, what it cannot do, and what can be certified about it. The answer to the first is that it is a Gibbs average with an additive sufficient statistic, which means it is a monoid and can be merged in any tree, and that it is neither idempotent nor homogeneous nor bilinear, with the last defect quantified at 2.1932. The answer to the second is a set of exact identities with matching error terms: the derivative in temperature is a covariance, the second derivative a variance, the third a third cumulant, the coarse-graining gap is an entropy divided by β, and finite sampling costs a total-variation distance that cannot be amplified by the logit parameterisation.

The answer to the third question is the result I would keep if I could keep only one. A fusion operator that carries a score alongside a value needs the normalising mass as well, not for numerical convenience but for information: with a value-only summary the merged value is undetermined on the convex hull of the shard values, and no amount of cleverness recovers it, because the construction is a convex combination and the combination's weights are exactly the information that was discarded. The minimax rule over that uncertainty is the midpoint, whose error is half the range and is attained, and which is never beaten by equal weighting in the instances tested. This is not a theorem about a software bug; it is a theorem about the operator, and it converts a silent, systematic error into a stated interval.

The answer to the fourth is a short list of certified statements, each with a condition of validity that is written down rather than assumed: an error budget for sampling and truncation composed into one number, a temperature calculus that tells an analyst what changes and by how much when β moves, and a quantum reading in which the classical theory is the commuting case and coherence is provably not a rounding error. Two of those statements are negative — the certificate saturates, the coverage falls short — and they are in the paper because a budget that is only trusted when it looks good is not a budget.

What remains open is worth naming so the paper does not pretend to completeness. The sharp constant in the quantised ledger is not known. The composition of a certificate with an estimated temperature is missing. The empirical question of how often real aggregation weights fall outside the δ for which quantised merging is safe has not been measured on production data, only argued from a single published tolerance study. And the minimax statement is one-dimensional in its sharp form; the vector-valued version is bounded by the diameter but is not sharp, and a practitioner with a vector aggregate will want the sharp version. Each of these is a question this paper makes precise rather than one it answers.

If there is a single sentence to carry away it is that the mass is not bookkeeping. It is the part of the record that makes the fusion invertible, and a system that drops it is not summarising the data it processed, only the values it happened to touch. The impossibility theorem in Section IV is the formal content of that sentence, and the ledger in Section III is the smallest object that satisfies it.

Appendix A. Proofs

Proofs are given for the results whose arguments are not a one-line citation. Notation is that of Section II: records $i = 1,\dots,n$ with scores w_i, values v_i, weights $p_i = e^{\beta w_i}/Z$, $Z = \sum_i e^{\beta w_i}$, aggregate value $V = \sum_i p_i v_i$ and aggregate score $W = \beta^{-1} \ln Z$.

A.1 Theorem 1 (ii) and (v): the merge laws

(ii). Write $S = \sum_i e^{\beta w_i} v_i$. For two disjoint record sets A, B the partition functions and values sums add, $Z_{A\cup B} = Z_A + Z_B$ and $S_{A\cup B} = S_A + S_B$, because the defining sums are over disjoint index sets. Hence $V_{A\cup B} = (S_A+S_B)/(Z_A+Z_B)$, which is the weighted mean of V_A and V_B with weights Z_A, Z_B. Commutativity and associativity follow because addition in the two components is commutative and associative. A formal neutral element would be (Z, S) = (0, 0), that is $W = -\infty$ with an empty record set; no non-empty record set attains it, because $Z > 0$ whenever a record exists. The structure is therefore a commutative monoid whose identity is not an element of the carrier set, and it is not a group. □

(v). Fix a record set and consider the map $\beta \to (W, V)$. By the previous part the pair is additive over disjoint unions and determined by n + 2 scalars, so it is a sufficient statistic of fixed dimension in the sense of exponential families [18]. The sufficiency is with respect to the family obtained by reweighting records by $e^{\beta w}$, which is exactly the family in which the data enter. □

A.2 Theorem 3: the reachable set is the convex hull

Let there be k blocks with reported pairs (V_j, n_j), $j = 1,\dots,k$, and let V be the true global aggregate. Because $V = \sum_j (Z_j/Z) V_j$ with $Z = \sum_j Z_j$ and $Z_j > 0$, every attainable V is a convex combination of the reported values, so the reachable set is contained in $\mathrm{conv}\{V_1,\dots,V_k\}$.

For the reverse inclusion let m lie in the relative interior of the simplex and put $Z_j = m_j Z$ for an arbitrary $Z > 0$. It remains to see that masses Z_j with the given V_j are realisable by actual records. Take a single record per block with value $v_j = V_j$; its partition function is $e^{\beta w_j}$, which equals any prescribed positive number for the score $w_j = \beta^{-1} \ln Z_j$. Hence every interior point, and by closure every boundary point, is attainable, and the reachable set is the hull. Note that the construction uses one record per block, so the undetermination is not a consequence of small blocks; it persists for blocks of any size. □

Remark. The argument shows exactly which extra scalar removes the ambiguity: the mass Z_j, or any strictly monotone transform of it. Two values plus two masses determine the merge; two values alone determine nothing beyond the hull.

A.3 Theorem 4: the minimax rule is the midpoint, and the constant is sharp

Write $a = \min_j V_j$ and $b = \max_j V_j$, so the reachable set is C = [a, b] by Theorem 3. For an estimate c the worst-case absolute error over C is $\max(b-c, c-a)$, which is minimised when the two terms are equal, that is at $c = (a+b)/2$, where the value is $(b-a)/2$. This is the Chebyshev centre of an interval and the argument is elementary. For sharpness, take the admissible masses $m = e_j$ concentrated on the block attaining a, and then on the block attaining b: both are attainable by A.2, and at them the error of the midpoint equals exactly half the range. Hence no rule does better in the worst case, and the midpoint does no worse. □

Remark. In more than one dimension the same argument gives the radius of the smallest Euclidean ball containing the hull, which is the Chebyshev radius and is not in general attained by a simple formula; the diameter bounds it and is not sharp. This is why the theorem is stated for scalar aggregates.

A.4 Theorem 6: the temperature calculus

Differentiating $\ln Z$ with respect to β gives $\sum_i p_i w_i$, and differentiating again gives $\sum_i p_i w_i^2 - (\sum_i p_i w_i)^2$, which is $\mathrm{Var}_p(w)$. The third derivative is the third cumulant of w under p by the same computation, since the cumulant generating function of w under p is $\ln \sum_i p_i e^{t w_i}$ and its derivatives at zero are the cumulants. For the value, differentiate $V = \sum_i p_i v_i$ using $\partial p_i/\partial\beta = p_i(w_i - \sum_l p_l w_l)$, which is the score's deviation from its mean. Collecting terms gives $dV/d\beta = \sum_i p_i v_i w_i - (\sum_i p_i v_i)(\sum_i p_i w_i) = \mathrm{Cov}_p(v, w)$. Convexity of $\beta \to \ln Z$ is $\mathrm{Var}_p(w) \geq 0$, and monotonicity of $\beta \to V$ is the sign of that covariance. □

A.5 Theorem 7: the coarse-graining gap is an entropy

Let the record set be partitioned into blocks B with masses $\pi_B = Z_B/Z$ and block scores $W_B = \beta^{-1} \ln Z_B$. The exact score of the union satisfies $\ln Z = \ln \sum_B Z_B$, so

$$W - \sum_B \pi_B W_B = \beta^{-1}[\ln Z - \sum_B \pi_B \ln Z_B] = \beta^{-1} H(\pi), \quad \text{(A.1)}$$

because $\sum_B \pi_B \ln \pi_B = \sum_B (Z_B/Z) \ln(Z_B/Z)$, and the rearrangement of the two logarithms gives the stated entropy. Non-negativity is Gibbs' inequality, and $H(\pi) \leq \ln k$ for k blocks with equality at uniform masses. Dividing by β gives the stated bound $\beta^{-1} \ln k$. □

Remark. The gap is an entropy and not a divergence, and the distinction is not cosmetic: a divergence would vanish when the block structure is trivial for every temperature, whereas $H(\pi)$ is a genuine information loss at every β and is independent of the values.

A.6 Theorem 8: the chain rule, and the singleton case

For a two-level partition with blocks A and then sub-blocks inside each, the Shannon chain rule gives $H(\pi) = H(\pi^{\mathrm{top}}) + \sum_a \pi_a H(\pi^{\mathrm{sub}}_a)$ [2, ch. 2]. Applying (A.1) at each level and adding shows that the coarse-graining gap is additive along the hierarchy, so a tree-shaped summary loses exactly the sum of the level entropies. If the finest level consists of singletons, then $H(\pi^{\mathrm{sub}}_a)$ is the conditional entropy of the record given its block, and the total gap is the entropy of the normalised weights, $\beta^{-1}H(p)$, which is the maximum information a summary that keeps only block aggregates can lose. □

A.7 Theorem 9: the logit parametrisation does not amplify error

Let p and q be weight vectors with q_i proportional to $p_i(1+\delta_i)$ and $|\delta_i| \le \varepsilon < 1$, which is what a relative error ε in the exponentiated scores produces. Write $N = \sum_i p_i(1+\delta_i)$, so that $|N-1| \le \varepsilon$ and hence $1/N \le 1/(1-\varepsilon)$. Then

$$\tfrac{1}{2}\|q-p\|_1 = \tfrac{1}{2}\textstyle\sum_i p_i|(1+\delta_i)/N - 1| \le \tfrac{1}{2}[(\sum_i p_i|\delta_i|)/N + |1/N - 1|] \le \varepsilon/(1-\varepsilon), \tag{A.2}$$

where the last step uses $1/N \le 1/(1-\varepsilon)$ and $|1/N-1| \le \varepsilon/(1-\varepsilon)$. The amplification factor is therefore one over one minus epsilon, which is close to one for small errors and diverges only as $\varepsilon \to 1$. □

A.8 Theorem 10: sampling and truncation compose

Let p and q be the true and the estimated weights, with $\mathrm{TV}(p,q) \le \eta$. Since $V(p) - V(q) = \sum_i (p_i - q_i)v_i$, which is a bounded linear functional of the difference, its modulus is at most $(\max_i v_i - \min_i v_i) \cdot \mathrm{TV}(p,q)$, after recentring the values, since the difference sums to zero. That gives the multiplier D equal to the range of the values, and the composition of two error sources by the triangle inequality, truncated at one because a probability is being bounded. Combining with (A.2) for the truncation of the score tail gives

$$|\hat{V} - V| \le D \cdot \min(1,\ \hat{\tau} + \varepsilon/(1-\varepsilon)), \tag{A.3}$$

with $\hat{\tau}$ the estimated sampling error and ε the truncation level. The minimum is taken because the right-hand side is a bound on a total-variation distance, which cannot exceed one. □

A.9 Lemma 2: diagonal states

For $\rho = \mathrm{diag}(p)$ and $\sigma = \mathrm{diag}(q)$, the difference is diagonal with entries $p_i - q_i$, so the trace norm is $\sum_i |p_i - q_i|$ and the first identity follows after halving. The spectral decomposition of a diagonal matrix is its entrywise decomposition, so the von Neumann entropy $-\mathrm{tr}(\rho \ln \rho)$ equals $-\sum_i p_i \ln p_i$. For the error probability, the Helstrom expression for two states with equal priors is $(1 - \tfrac{1}{2}\|\rho-\sigma\|_1)/2$ [5], which with the first identity becomes $(1-\mathrm{TV})/2$. □

A.10 Theorem 12, verified numerically

The counterexample of Section VIII is the pair of four-dimensional vectors $u = \tfrac{1}{2}(1,1,1,1)$ and $s = \tfrac{1}{2}(1,1,-1,-1)$, both of unit norm and mutually orthogonal. Their outer products have diagonal entries ¼ in every position, so the dephased states coincide exactly. Evaluating the trace distance of the dephased pair gives 0.0 to machine precision and of the pure pair gives 1.0 to machine precision, a difference of exactly one. The code that performs this check constructs the states from the vectors rather than from stored matrices, so the agreement is not a consequence of the stored artefact. □

Appendix B. Stored scalars

Every number quoted in the text is recoverable from the files below. The residuals are absolute deviations from the predicted identity in the units of the quantity concerned; the worst-case columns are maxima over the stated instance count.

Table B.1. Scalars quoted in the text and the file that stores them.

Quantity	Value	File
Master seed	2026110101	all
softmax reconstruction error, max over 400	2.2×10^{-16}	e1_representability.json
temperature identity residuals (three derivatives)	2.98×10^{-9}, 2.77×10^{-9}, 8.47×10^{-7}	e2_temperature_calculus.json
coarse-graining residual, max over 320	2.2×10^{-15}	e3_coarse_graining.json
sharp gap at $\beta = 1.5$ with eight blocks	$\ln 8/1.5 = 1.3862943611$	e3_coarse_graining.json
mean realised gap over 320 instances	1.3063	e3_coarse_graining.json
chain-rule residual; entropy identity residual	8.9×10^{-16}; 1.8×10^{-15}	e4_hierarchy.json
minimum excess in the pure-state inequality	8.6×10^{-5}	e5_quantum_boundary.json
coherence contribution to trace distance, mean	0.314	e11_dephasing.json
Helstrom floor: smallest slack over 400	1.2×10^{-5}	e6_helstrom.json
Helstrom floor: mean slack at $\beta = 8$	0.235	e6_helstrom.json
bootstrap coverage, eight cells	0.892–0.983	e7_sampling.json
saturated certificate at $\beta = 3$, $m = 10$	1.6267 against error 0.1017	e7_sampling.json
von Neumann–Shannon residual, max over 200	4.4×10^{-16}	e8_entropies.json
logit non-amplification: minimum slack	0.0197	e9_nonamplification.json
composed certificate: median ratio, max ratio	0.056, 0.308	e10_pipeline.json
dephasing: mean full, diagonal, coherence parts	$0.562 = 0.248 + 0.314$	e11_dephasing.json
hull fraction; max violation	1.000; -6.4×10^{-8}	e12_impossibility.json
worked example: two values and the gap	3.3333 vs 4.8786; gap 1.5452	e12_impossibility.json
minimax: max deviation from half the range	8.9×10^{-16}	e12_impossibility.json
equal weighting vs minimax: mean, max ratio	1.1277, 1.4634	e12_impossibility.json
idempotence defect; homogeneity defect at $\lambda = 2$	$\beta^{-1}\ln 2$; 2.1932	e13_binary_algebra.json
worked example at $\beta = 1$	$V = 10.5586$, $W = 2.2331$	e14_worked_example.json
$dV/d\beta$ at $\beta = 1$; effective record count	−0.108319; 4.812	e14_worked_example.json

Quantity	Value	File
skewness peak	17.8629 at β = 0.1543	e14_worked_example.json
quantisation: violations; max error/bound	0 of 600; 0.31	e15_quantisation.json
quantisation: mass chain at δ = 0.01, 0.1	0.00111, 0.01111	e15_quantisation.json
tree merging: max relative value gap; score gap	3.7×10^{-16}; 0	e16_tree_merge.json

Appendix C. Repository manifest

Table C.1. Contents of the deposit. The SHA-256 manifest is authoritative for what is present; this table says what each path is for.

Path	Contents
otaz_quantum/core.py	the reference implementation: primitives, the binary and n-ary algebra, the temperature calculus, coarse-graining and hierarchy, the ledger (impossibility, minimax, quantisation, tree merging), the quantum interface, sampling
experiments/ otaz_run_quantum.py	the sixteen experiments, seeded, writing nineteen JSON files
tests/test_core.py	fifty-two unit tests, including seven for the ledger statements
scripts/ otaz_verify_quantum.py	fifty-six independent checks; does not import the library
scripts/ otaz_checksums.py	writes and verifies the SHA-256 manifest
otaz_make_figures.py	the eight figures, in SVG and PNG, black line art only
otaz_make_cover.py	the deposit cover sheet
paper/	this manuscript as HTML plus the builder, the two style sheets and the section modules sec_1_front to sec_8_appendix
results/	nineteen JSON files, each with the property, the instance count, the residuals and a verdict
figures/	fig1–fig8 in SVG and PNG
publication/	the built PDFs, the cover sheet, the one-page Arabic novelty assessment (OTAZ_III_Novelty_Assessment_AR, with its HTML source and build script), metadata and the deposit archives
docs/	reading guide, release steps and the colour policy
README.md, CHANGELOG.md, CITATION.cff	orientation, version history and citation record

Licensing: the text and the results are released under CC BY-NC 4.0; the code is released under the PolyForm Noncommercial License 1.0.0. Neither permits commercial use, which is deliberate. The repository contains no third-party data.

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